Rethinking the Evaluation of Harness Evolution for Agents

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HarnessRSI 63 ★★

evidence: task

Paper

Code

1 What is being tested?

Critique, not a universal refutation. Names Meta-Harness, AHE and AEVO; experimentally instantiates **AHE with external exploration disabled**, not independent reruns of all three.

1 Fix weights; allocate extra search

Same initial harness on Terminal-Bench.
Feedback: trajectories; outcomes if available.

2 Choose what is allowed to change

Parallel: independent attempts; harness fixed.
Sequential: revise solutions; harness fixed.
Evolution: one shared harness across tasks.
Scaling: a separate harness for each task.

3 Read feedback; edit or retry

Harness edits: prompt, tools, middleware, memory.
Baselines also get the regime's task feedback, but cannot edit their harness.

4 Select; distinguish the claim

No tests: last sequential/harness iterate; parallel uses model self-selection.
Tests: outcome-based selection; held-out experiment selects harness on validation.

Two different goals: reusable shared-harness improvement vs same-task adaptation. Positive game results test the latter.

Game adoption: ARC exposes update tools during play. Edge pauses for edits after three submissions or 120 minutes; validated changes are installed, otherwise the old harness stays. The validation criterion is unspecified.

2 Terminal: shared gains do not transfer

Terminal-Bench 2.1 · K = 5 · model-average scores (%)

Method	No tests pass@1	With tests pass@1 / pass@5
Initial harness	68.2	67.6 / —
Parallel Sampling	**72.3**	**82.0 / 82.0**
Sequential Refinement	69.3	76.9 / **87.1**
Harness Evolution	**67.4**	69.7 / 79.0
Harness Scaling	71.8	75.5 / 85.6

Tables 1–2: Claude Opus 4.6, GPT-5.4, GPT-5.4 mini; 89 tasks. Two-run means, no SD/CI. pass@1 = mean rollout reward; pass@5 = any success. Compare within each table.

Held-out: 63.2% → 63.2%
Table 3: 45 train / 10 validation / 34 test tasks. Validation selects shared harness. No average test gain.

3 Games: same-task adaptation helps

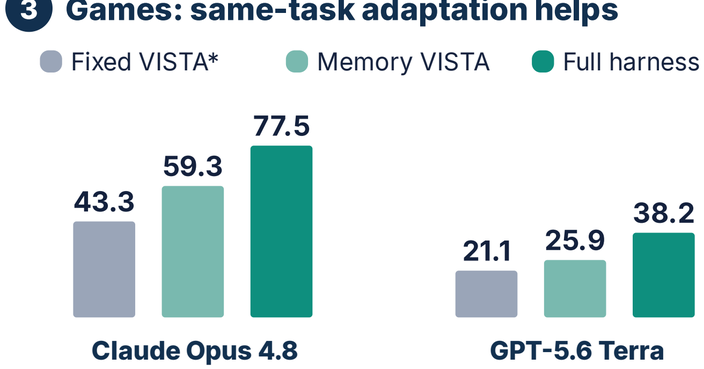


Table 4: RHAE, not win %. All 25 public ARC-AGI-3 games; 2,000 environment actions. ARC run count unspecified; no SD/CI.

EdgeBench game subset · 12-hour best score

Table 5 · 0–100	Fixed	Scaling
Claude Opus 4.8	48.3	**54.7**
GPT-5.4	45.9	**49.8**

Five bot-programming tasks only. Two independent runs, no SD/CI. Not held-out-game transfer.

4 Controls and scope limits

- **Budget ≠ one universal resource:** Terminal matches rollouts, excludes editing cost. ARC matches actions, not internal reasoning. Edge includes editing and pauses in 12 hours.
- **ARC access:** VISTA* disables memory files; VISTA restores evolving memory; full adds skills/tools/hooks. No isolated component or equal-write-access update-rule ablation.
- **Selection ≠ tuning:** Terminal validation selects the harness; defaults come from a reference snapshot. ARC settings avoid saturation. No separate baseline tuning budgets or tuning splits reported.
- **K = 10 exception:** with tests, Claude Opus 4.6 Scaling 90.5% / 94.4% vs Sequential 89.6% / 93.1% (pass@1 / pass@10, Table 8). Authors call gap within variance; none printed.
- **Scope:** no held-out-game or cross-model artifact transfer, no improved-improver test. No paired-seed protocol; two-run means are not variance estimates.

5 Where improvement is absent

- **Model dependence:** no-test GPT-5.4 shared evolution 75.3 → 69.7%; Claude Opus 4.6 task-specific Scaling leads 76.0 vs Parallel 74.7% (Table 1).
- **ARC is not uniform:** Claude Opus 4.8 tn36: VISTA* 18.5 → full 3.6 RHAE (Table 4).
- **Edge is uneven:** Claude Opus 4.8 NetHack 44.3 → 61.3, but Wesnoth 88.0 → 86.0. GPT-5.4 OpenRCT2 36.2 → 50.0, NetHack 50.4 → 49.0 (Table 5; score points).

Small task differences have no uncertainty estimates. The paper does not establish their statistical reliability.

Not established: exact total-compute matching; separate baseline tuning budgets; an improved improvement procedure. Missing numerical settings include Terminal temperature and split seed, and an ARC internal token ceiling.

Edge reports the best score across visible submissions and evaluator-only snapshots, not a fresh held-out-game test.

Takeaway

Separate reusable improvement from adaptation to the scored task. Compare against feedback-matched search, then test the retained harness on unseen tasks. Games show useful adaptation—not a general transfer result.

Our read · HarnessRSI

Score 63 · ★★ · evidence: task
Reports: held-out diagnostic, budget controls, artifact variants.
Does not report: SD/CI, game transfer, full optimizer costs.
For our board: qualify budget, leakage and ablation labels; a flat held-out test is not positive retained evidence.